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ORIGINAL RESEARCH

Utility Scores In Patients With Oxygen-Dependent COPD: A Case-Control Study

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Abstract

Background: Utility measures that summarize the health-related quality of life of an individual using a single number usually between 0 (death) and 1 (full health) are useful to quantify the benefits of health care interventions in terms of quality-adjusted life years (QALYs) and to help prioritizing limited health resources from cost-effectiveness analyses among patients with different health conditions. **Objective:** To determine utility scores in patients with oxygen-dependent chronic obstructive pulmonary disease (COPD). **Methods:** Patients with oxygen-dependent COPD (the cases) were matched, on a 1:2 basis, to COPD controls according to gender, age (± 5 years) and FEV₁ ($\pm 5\%$ predicted). Utility scores were obtained from the SF-6D, a measure derived from the SF-36. **Results:** From a cohort of 102 patients with oxygen-dependent COPD, 68 (42 men; mean age: 71 years; mean FEV₁: 35% predicted) were successfully matched with 136 controls. We found clinically and statistically significant differences in mean utility scores between cases (0.588 ± 0.071) and controls (0.627 ± 0.085 ; $p = 0.001$). The same differences were observed in men and women. **Conclusion:** Oxygen-dependence adds to the burden of disease in terms of quality of life. These utility scores may be useful in cost-utility analyses involving patients with oxygen-dependent COPD.

Introduction

Quality of life (QoL) may be measured by generic or disease-specific questionnaires (1). In clinical trials, disease-specific questionnaires that focus on specific area of QoL [such as the Chronic Respiratory Questionnaire (2) or the St. George's Respiratory Questionnaire (3)] are preferred because they are more likely to detect changes in quality of life. Generic questionnaires are designed to measure all important aspects of QoL and are useful in comparing health-related QoL among patients with different chronic health conditions.

Generic instruments include health profiles [such as the Medical Outcome Survey - Short Form (SF-36) (4) and the Sickness Impact Profile (5)] and utility measures [such as the Health Utilities Index (6) or the Quality of Well-Being Scale (7)]. Utility measures express the health-related quality of life of an individual using a single score (often referred to as "quality-of-life weight") that incorporates the overall assessment of health-related QoL and the values attached to it. This number is typically between 0 and 1. The two extreme values are anchored to specific states, most commonly death (0) and full health (1). Utility scores may also be directly obtained using preference-based techniques [such as the time trade-off (8)]. A large number of studies have been conducted in order to determine the utility scores in a wide variety of health conditions,

Keywords: Case control study, Chronic Obstructive Pulmonary Disease, oxygen dependent, utility scores

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including chronic obstructive pulmonary disease (COPD) (9). These scores have been compiled in health catalogs or registries of health state utility weights for use in cost-utility analysis (10–12).

The QoL weight of patients with oxygen-dependent COPD remains unexplored however. This information would be important not only to compare the burden of the disease with other health conditions, but also to generate cost-effectiveness (i.e., cost-utility) analyses and to better inform resources allocation. The objective of this study was to determine the utility scores of patients with oxygen-dependent COPD, and to compare them with those of patients with non-oxygen dependent COPD. We hypothesized that for equally severe airflow obstruction, patients with oxygen-dependent COPD have lower utility scores than patients with non-oxygen-dependent COPD.

Methods

Study population

This study took place in the province of Quebec, Canada.

Cases

Patients with oxygen-dependent COPD were identified from the Quebec City area respiratory home care program. This service is affiliated to our hospital and delivers home care (mainly long-term oxygen therapy and related services) to all COPD patients living in the area and meeting the usual criteria for long-term oxygen therapy (13). These services are covered by the provincial government, through the Quebec provincial universal medical insurance plan. Accordingly, the registered patients are not selected on any clinical, social or financial basis.

Outpatients with a clinical diagnosis of COPD, supported by a history of past or current smoking and obstructive disease with a forced expiratory volume in 1 second (FEV_1) $\leq$ 50% of predicted value and a FEV_1 /force vital capacity (FVC) $<$ 70%, who were registered at the respiratory home care program were asked by a visiting nurse or respiratory therapist to participate in this study. Oxygen dependence was defined as severe hypoxemia (resting daytime $PaO_2 \leq 55$ mmHg or ≤ 59 mmHg in the presence of at least one of the following: (1) peripheral edema; (2) right ventricular hypertrophy on ECG; (3) hematocrit $\leq 55\%$) requiring continuous oxygen therapy (i.e., ≥ 18 hours per day). Informed consent was obtained prior to the administration of the questionnaires.

Controls

Controls were drawn from a cohort of 409 patients with COPD who participated in a longitudinal study taking place in the province of Quebec and looking at the natural history of the disease. This longitudinal study was a precursor of the CanCOLD study (14). At study entry, demographic and medical characteristics were collected in all patients who were also administered the SF-36.

Patients with oxygen-dependent COPD (the cases) were matched, on a 1:2 basis, to controls according to gender, age (± 5 years) and FEV_1 ($\pm 5\%$ predicted). If more than 2 controls were available for one case, then the 2 patients who best matched the case were selected. Cases who could not match with 2 controls were excluded. Patients could serve as controls only once.

Measurement instruments

We measured generic QoL by administering the SF-36 (4). The SF-36 is a self-completed questionnaire that measures 8 dimensions of health: physical functioning (10 items), role limitation due to physical problems (4 items), role limitation due to emotional problems (3 items), social functioning (2 items), mental health (5 items), energy/vitality (4 items), bodily pain (2 items) and general health perceptions (5 items). The SF-36 items and scales are scored so that a higher score indicates a better health state. For comparison, we used data obtained from an age-matched Canadian population (15). Utility scores were obtained from the SF-6D (16), a utility measure derived from the SF-36. The SF-6D was specifically developed in order to “reconcile a profile health status measure, the SF-36, with the quality-adjusted life-years” approach (16). In this validation study, SF-36 items were valued using a visual-analogue scale and from standard gamble and then mapped into a 0.29 to 1.00 scale (1.00 indicating full health). Although different methods for estimating the minimal clinically important difference (MCID) in generic utility-based measures lead to varying conclusions (17), the best estimate for the MCID of the SF-6D is 0.033 (18).

Other characteristics were collected and included the following: (1) age; (2) sex; (3) the latest available pulmonary function testing (FEV_1 , FEV_1/FVC) and arterial blood gas results that were collected within the 6 months preceding the questionnaires administration; (4) previous psychiatric disorders and past medical history (by considering 6 diagnostic categories: heart failure, ischaemic heart disease, diabetes, stroke, musculoskeletal disease, and cancer); (5) current medication; and (6) time since the prescription of long-term oxygen therapy when appropriate. Disease severity was defined according to the GOLD classification of airflow limitation (mild: $FEV_1 \geq 80\%$ predicted; moderate: $50\% \leq FEV_1 < 80\%$ predicted; severe: $30\% \leq FEV_1 < 50\%$; very severe: $FEV_1 < 30\%$ predicted) (19).

Statistical analysis

Descriptive statistics (proportions, median and interquartile range, or mean, standard deviations and associated 95% confidence intervals (95% CI) when appropriate) were used to describe the study population. Matched sets without replacement of case and control subjects (2:1 matching) were formed. For categorical variables, conditional logistic univariate analysis appropriate for paired data was used for the comparisons between groups. For continuous variables, a two-way

ANOVA with blocking factor linked to subjects was performed. Statistical significance was present when the two-tailed p value <0.05 .

Results

Patients characteristics

One hundred and seventy-six (176) patients with oxygen-dependent COPD were registered at the Quebec City area respiratory home care program. Of those, 74 did not fill in the SF-36 for the following reasons: 27 refused to participate; 12 were not invited to participate in due time; 10 were mentally inept; 16 did not provide sufficient data for analysis; 8 were unavailable at the time of the study; and 1 was not fluent in French. Therefore, 102 patients were surveyed and completed the SF-36. From the 102 patients who did complete the SF-36, 68 were successfully matched with 2 controls (total: 136 controls) on the basis of gender, age and FEV₁. Table 1 summarizes the characteristics of the cases and controls. Both groups were similar not only with regard to the matching variables, but also according to the burden of diseases. On average, patients had severe COPD.

SF-36 and utility (SF-6D) scores

The results obtained from the SF-36 administration are presented in Table 2. These results are best understood by comparing them to data obtained from an age-matched population ("Normative data" column in Table 2).

Table 1. Baseline clinical characteristics

Characteristics	Cases (n = 68)	Controls (n = 136)
Gender (male, %)	42 (62%)	84 (62%)
Age (years, mean, SD)	71 (7)	68 (8)
FEV ₁ (% pred, mean, SD)	35 (14)	37 (12)
Severity of COPD		
Very severe (n, %)	28 (41%)	45 (33%)
Severe (n, %)	29 (43%)	69 (51%)
Moderate (n, %)	10 (15%)	21 (15%)
Mild (n, %)	1 (1%)	1 (1%)
PaO ₂ (mm Hg, mean, SD)	54 (9)	72 (9)
PaCO ₂ (mm Hg, mean, SD)	53 (9)	43 (6)
Past medical history (number of major medical diagnoses, excluding COPD) ^a		
0	41 (59%)	81 (60%)
1	23 (34%)	39 (29%)
2	3 (4%)	11 (8%)
≥3	1 (1%)	5 (4%)
Number of drug prescriptions (median, min-max)	7 (2–16)	8 (0–31)
Time since the introduction of long-term oxygen therapy (months, mean, SD)	26 (25)	N/A

^aThe following diagnostic categories were considered: heart failure, ischaemic heart disease, diabetes, stroke, musculoskeletal disease, cancer, and depression; N/A: not applicable.

Table 2. SF-36 results, mean (SD)

	Cases (n = 68)	Controls (n = 136)	P value	Normative data ^a
Physical component	26 (7)	33 (7)	<0.0001	51
Physical functioning	23 (22)	34 (19)	0.002	86
Role—physical	8 (9)	51 (27)	<0.0001	82
Bodily pain	59 (30)	71 (24) ^a	0.006	76
General health	33 (17)	49 (20)	<0.0001	77
Mental component	40 (11)	49 (12) ^b	<0.0001	52
Vitality	44 (20)	54 (19)	0.0005	66
Social functioning	57 (31)	71 (26)	0.0008	86
Role—emotional	15 (11)	71 (26)	<0.0001	84
Mental health	62 (20)	69 (20)	0.03	77

^aAge- (65–74) and sex-standardized scores for the 8 domains of the SF-36 and its 2 components; from reference (15).

All comparisons with cases and controls: $p < 0.001$, with only 2 exceptions: a: $p = 0.038$; b: $p = 0.024$.

We found significant impairments in all domains of health-related QoL in both cases and controls. In all SF-36 domains, the scores were even lower in patients with oxygen-dependent COPD. Also, all comparisons with the normative data (in both cases and controls) were statistically significant.

Utility scores are summarized in Table 3. The mean utility score was 0.588 (SD: 0.071) in oxygen-dependent COPD patients and 0.627 (SD: 0.085) in those with non-oxygen-dependent COPD. The difference was clinically and statistically significant, by exceeding the MCID of 0.033. Similar results were obtained in both men (difference between cases and controls: 0.036) and women (difference between cases and controls: 0.045).

Discussion

We found that QoL of patients with oxygen-dependent COPD is profoundly impaired. For comparison, the utility scores we found in severe oxygen- and non-oxygen-dependent COPD are worse than those associated to a large myocardial infarction, stroke leaving permanent moderate deficit, or dissecting or ruptured aortic aneurysm (12). We also confirmed our hypothesis that, for equally severe airflow obstruction, patients with oxygen-dependent COPD have lower utility scores than patients with non-oxygen-dependent COPD. Because long-term oxygen therapy is usually provided by a stationary oxygen concentrator and is recommended to be used for at least 15–18 hours a day, such a therapeutic regimen limits the ability of the patients to remain active and may

Table 3. Utility scores, mean (SD)

	Cases	Controls	P value
All patients (68 cases: 136 controls)	0.588 (0.071)	0.627 (0.085)	0.001
Men (42 cases: 84 controls)	0.598 (0.069)	0.634 (0.080)	0.013
Women (26 cases: 52 controls)	0.572 (0.073)	0.617 (0.093)	0.034

be detrimental to the rehabilitation process. The contribution of the limitations imposed by the concentrator cannot be dissociated from the total burden of oxygen dependence.

Utility scores can be used in the evaluation of the cost-effectiveness of health care interventions. Such evaluations express the benefits of the interventions in terms of quality-adjusted life-years (QALYs) (20). The product of the utility score and the duration of life will give the QALYs. From our finding of utility scores around 0.6, oxygen-dependent COPD reduces quality of life by about 40%. The disease therefore takes away 0.4 QALY over a one-year period. In cost-utility analyses, costs per QALY are considered. For instance, in a hypothetical trial of a new bronchodilator in COPD, if a 2-year study found that, on average, \$1500 must be added to the cost of usual care to improve quality of life by 0.05 QALY, then the incremental cost-effectiveness ratio would be \$30 000 (i.e., \$1500 / 0.05) per QALY gained over 2 years. For health care payers and consumers, such information can only be interpreted in the context of the cost-effectiveness of other interventions (10).

No gold standard exists for utility measurement. Several head-to-head comparisons between utility measures have been published (21–23), sometimes in asthma and COPD (16, 24). Most studies suggest that the available instruments are not totally interchangeable because they are scaled differently and produce varying results (22). Most measures have agreed with known differences in health, supporting their validity when used, as we did, as discriminative instruments in COPD (Table 4) (24–27). However, the differences in responsiveness (i.e., the extent to which an instrument can capture changes in quality of life over time (28)) and the inequality in the instruments' MCIDs (29) are of greater concern in cost-utility analyses conducted from randomized trials, where utility scores (and associated QALYs) are used as denominators in cost-effectiveness ratios (30).

We undertook this study as a preliminary step to validate the primary outcome of the ongoing International Nocturnal Oxygen (INOX) trial, a 3-year,

multi-center, placebo-controlled, randomized trial of nocturnal oxygen therapy in COPD (ClinicalTrials.gov: NCT01044628). The INOX primary outcome is a composite of (1) all-cause mortality, or (2) requirement for LTOT. A composite outcome may be inappropriate and misleading when there is an important gradient between its components in terms of their respective importance to patients (31). We therefore wished to examine whether the composite of death and requirement for LTOT is appropriate by exploring the gradient of importance between these two components from the patients' perspective. To do so, we used the same methodology as in a systematic review of cardiovascular trials published in 2002–2003 (32).

In this systematic review, the authors first developed a hierarchical categorization of importance to patients for the components end points included in the eligible studies. Published estimates of utility associated with the outcomes guided this process (Table 5). Five categories of end points were hence created: death, critical, major, moderate and minor. A large gradient between components of a composite outcome was considered to be present when the composite outcome included components from categories I or II (fatal or critical) with components from category V (minor). A moderate gradient was considered to be present if the components of the composite outcome were from categories I or II (fatal or critical) and from category IV (moderate).

Minor (or absent) gradients were those not included in the other two categories. From 114 trials, only 14 (12%) included a composite outcome with either no gradient or minor gradient in importance to patients. The pooled treatment effects of interventions on fatal and critical outcomes were similar. The authors concluded that confident interpretation of composite outcomes first requires small gradients of importance to patients. Applying the same methodology to the INOX trial composite outcome, we determined that the prescription of LTOT would be classified as "critical" (Table 5). Accordingly, the gradient of importance to patients between death and prescription of LTOT is minor. Based on this, we therefore consider the composite of death and requirement for LTOT as appropriate.

Table 4. Selected examples of studies reporting utility scores in COPD

Measurement instrument	References	Setting	GOLD stages			
			Mild	Moderate	Severe	Very severe
EQ-5D	Rutten-van Molken (25)	Secondary analysis of the UPLIFT trial (randomized trial of tiotropium in COPD)	-	0.850	0.795	0.683
EQ-5D	Asukai (26)	Secondary analysis of the randomized trials of indacaterol in COPD	0.82	0.80	0.77	0.74
EQ-5D	Szende (24)	Population-based study in Northern Sweden	0.85	0.73	0.74	0.53
SF-6D	Szende (24)	Population-based study in Northern Sweden	0.80	0.73	0.73	0.62
Health Utility Index (HUI3)	Scharf (27)	Study of sleep in COPD	0.40	0.58	0.53	0.39

Table 5. Component outcomes by category of importance to patients

End point	Utility score ^a
I. Death	0.0
All cause mortality	
Mortality not otherwise defined	
Cardiac death not otherwise defined	
Cardiac death due to coronary heart disease not otherwise defined	
II. Critical	0.7–0.8
Large myocardial infarction	
Stroke leaving permanent moderate deficit	
Cardiac arrest followed by resuscitation manoeuvres	
Dissecting or ruptured aortic aneurysm	
III. Major	0.8–0.9
Non-fatal myocardial infarction	
Stroke leaving permanent deficit, severity not defined	
Coronary artery bypass grafting	
Cerebrovascular event not otherwise defined	
IV. Moderate	0.9–0.95
Coronary revascularisation not otherwise specified	
Coronary revascularisation—angioplasty/stenting	
Non-fatal angina needing hospital admission	
Hospital admission not otherwise specified	
V. Minor	
Non-fatal angina, not defined	>0.95
Non-fatal arrhythmias, not otherwise specified	
Dyspnoea, not otherwise defined	

Table adapted from (32), with permission.
^aUtility scores from reference (12).

Our study has both strengths and limitations. Since it was conducted in a jurisdiction with a universal medical insurance plan, we included patients who were not selected on any clinical, social or financial basis. Clinical assessment was straightforward. Although it allowed the detection of statistically and clinically important differences between cases and controls, the sample size was small. In the determination of the utility scores, the exclusion of a number of oxygen-dependent patients, with the consequent selection bias, was in part limited by the careful selection of controls.

Conclusion

We conclude that few diseases are associated with worse quality of life than oxygen-dependent COPD. The prescription of LTOT represents a major event in the life of patients with severe and very severe COPD. This study also provides reassurance regarding the validity and appropriateness of the primary outcome of the INOX trial. However, the use of utility scores in clinical trials and cost-effectiveness analyses still need further investigation.

Declaration of Interest Statement

The authors have no conflict of interest to declare relevant to the contents of the article for any of the five authors.

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